AUTOMATE INDIA

Problem Statement: Smart Waste Management Ecosystem

Route Optimization Theme: Smart City & IoT

Category: Hardware + Software Integration

Team Name: ByteFlow

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IDEA TITLE

- Autonomous Waste Segregation & Fleet Optimization System
- Core Function: Automatically segregates wet/dry waste at the source and prevents overflowing street bins.
- Features: Live dashboard tracking capacity and calculating shortest paths for garbage trucks.
- Innovation: Replaces expensive manual sorting with low-cost, automated edge-hardware.
- Uniqueness: Bridges physical IoT sensors with Data Structures & Algorithms (Greedy Approach) for dynamic routing.

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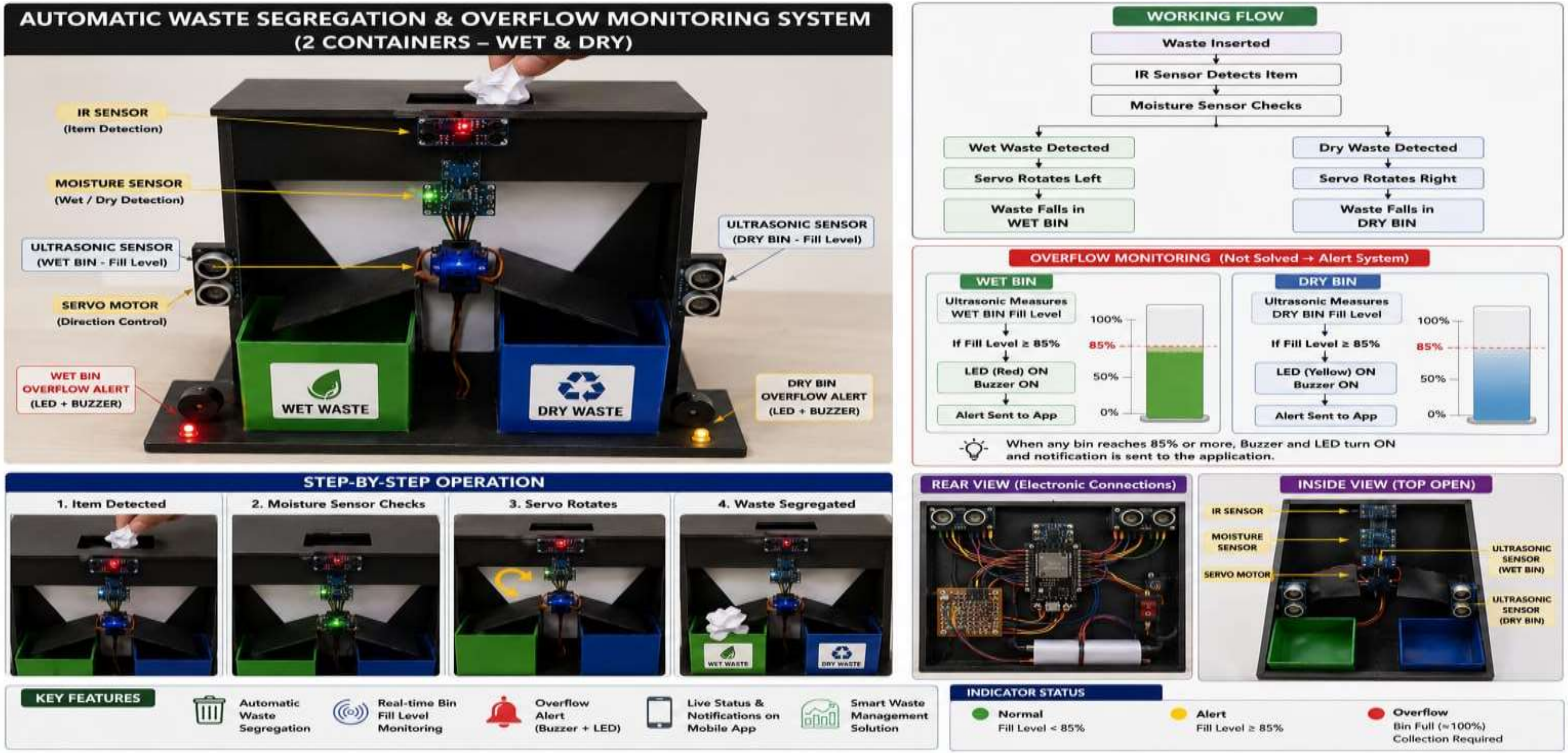
TECHNICAL APPROACH



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FEASIBILITY AND VIABILITY

- Feasibility
- Built using highly affordable, readily available components (Under ₹1,500 per unit).
- Modular design allows easy scaling across residential and commercial sectors. Challenges
- Sensor degradation (dirt on moisture sensor) over time.
- Wi-Fi/Network dropouts in dense urban pockets. Solutions
- Future upgrade to ESP32-CAM (AI Vision) for contactless classification.
- Local fail-safes: LEDs and buzzers operate entirely offline to prevent overflow if the network drops.

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IMPACT AND BENEFITS

- Target Audience

• Municipal Corporations, Sanitation Workers, and Urban Planners. Benefits

• 30% reduction in municipal fleet fuel consumption (Diesel savings).

• 100% source-level segregation, drastically improving recycling plant efficiency. Wider Impact

• Directly aligns with Swachh Bharat Abhiyan (Clean India Mission).

• Significantly reduces urban carbon footprint (estimated 2,400 tons CO_2 saved annually).

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RESEARCH AND REFERENCES

- Research Sources

• CPCB (Central Pollution Control Board) Guidelines on Solid Waste Management.

• IEEE papers on IoT-based Smart Dustbins and Graph Algorithms in Logistics.

• Local news reports detailing the mixed waste crisis in Greater Noida. Similar Solutions Studied

• Traditional static-route municipal garbage trucks (Found to be highly inefficient).

• Standard market "Smart Bins" (Found to only monitor capacity, lacking auto-segregation and fleet routing integration).